Do the Data Fit Mendel?

Using the χ^2 test to evaluate genetic crosses

Example Student

BIOL 109 · The Community College of Baltimore County

Why do we need a statistical test?

- Mendel's laws predict **ratios** of offspring, like 3:1 or 9:3:3:1
- Real crosses almost never match those ratios exactly
- So how far off is *too* far off?

The **chi-square (χ^2) goodness-of-fit test** tells us whether the difference between what we **observed** and what we **expected** is small enough to be explained by chance.

Background: where expected ratios come from

A monohybrid cross of two heterozygotes (Pp × Pp):

Punnett square for Pp × Pp

	P	**p**
P	PP	Pp
p	Pp	pp

- Genotypes: 1 PP : 2 Pp : 1 pp
- Phenotypes: **3 dominant : 1 recessive**
- A dihybrid cross (PpSs × PpSs) gives **9 : 3 : 3 : 1**
- These ratios were first reported by Mendel (1866)

Hypotheses

Every χ^2 test starts with two competing statements:

- **Null hypothesis** (H_0): the observed counts fit the expected Mendelian ratio; any differences are due to random chance.
- **Alternative hypothesis** (H_1): the observed counts do **not** fit the expected ratio.

We choose a significance level before looking at the data: $\alpha = 0.05$.

The χ^2 statistic

$$\chi^2 = \sum_{i=1}^{k} \frac{(O_i - E_i)^2}{E_i}$$

- O_i = observed count in category i
- E_i = expected count in category i = total × expected proportion
- k = number of categories

Degrees of freedom: $df = k - 1$

Critical values

If χ^2 is greater than the critical value, we **reject** H_0.

```
data.frame(df = 1:4, critical_value = round(qchisq(0.95, df = 1:4), 2))
```

A data.frame: 4 × 2

df	critical_value
<int>	<dbl>
1	3.84
2	5.99
3	7.81
4	9.49

Critical values at $\alpha = 0.05$, computed in R with `qchisq()`. These match the table in Chapter 11 of the lab manual.

Example 1: monohybrid cross (3:1)

Data from the Chapter 11 worked example: 78 tall and 22 short plants.

```r
observed <- c(Tall = 78, Short = 22)
ratio    <- c(3, 1)
expected <- sum(observed) * ratio / sum(ratio)
data.frame(observed, expected = round(expected, 1),
           contribution = round((observed - expected)^2 / expected, 3))
```

A data.frame: 2 × 3

	observed	expected	contribution
	<dbl>	<dbl>	<dbl>
Tall	78	75	0.12
Short	22	25	0.36

```r
chisq.test(observed, p = ratio / sum(ratio))
```

```
    Chi-squared test for given probabilities

data:  observed
X-squared = 0.48, df = 1, p-value = 0.4884
```

Example 1: interpretation

- $\chi^2 = 0.48 < 3.84$ (critical value, $df = 1$)
- $p = 0.49 > 0.05$

Conclusion: we **fail to reject** H_0. The data are consistent with a 3:1 ratio; the small differences are likely due to chance.

Example 2: dihybrid cross (9:3:3:1)

Data from the Chapter 11 worked example: 320 seeds.

```
observed <- c(`Round yellow` = 190, `Round green` = 65,
              `Wrinkled yellow` = 40, `Wrinkled green` = 25)
ratio    <- c(9, 3, 3, 1)
expected <- sum(observed) * ratio / sum(ratio)
contribution <- (observed - expected)^2 / expected
data.frame(observed, expected, contribution = round(contribution, 3))
```

A data.frame: 4 × 3

	observed	expected	contribution
	<dbl>	<dbl>	<dbl>
Round yellow	190	180	0.556
Round green	65	60	0.417
Wrinkled yellow	40	60	6.667
Wrinkled green	25	20	1.250

Example 2: the test

```
1 di_result <- chisq.test(observed, p = ratio / sum(ratio))
2 di_result
```

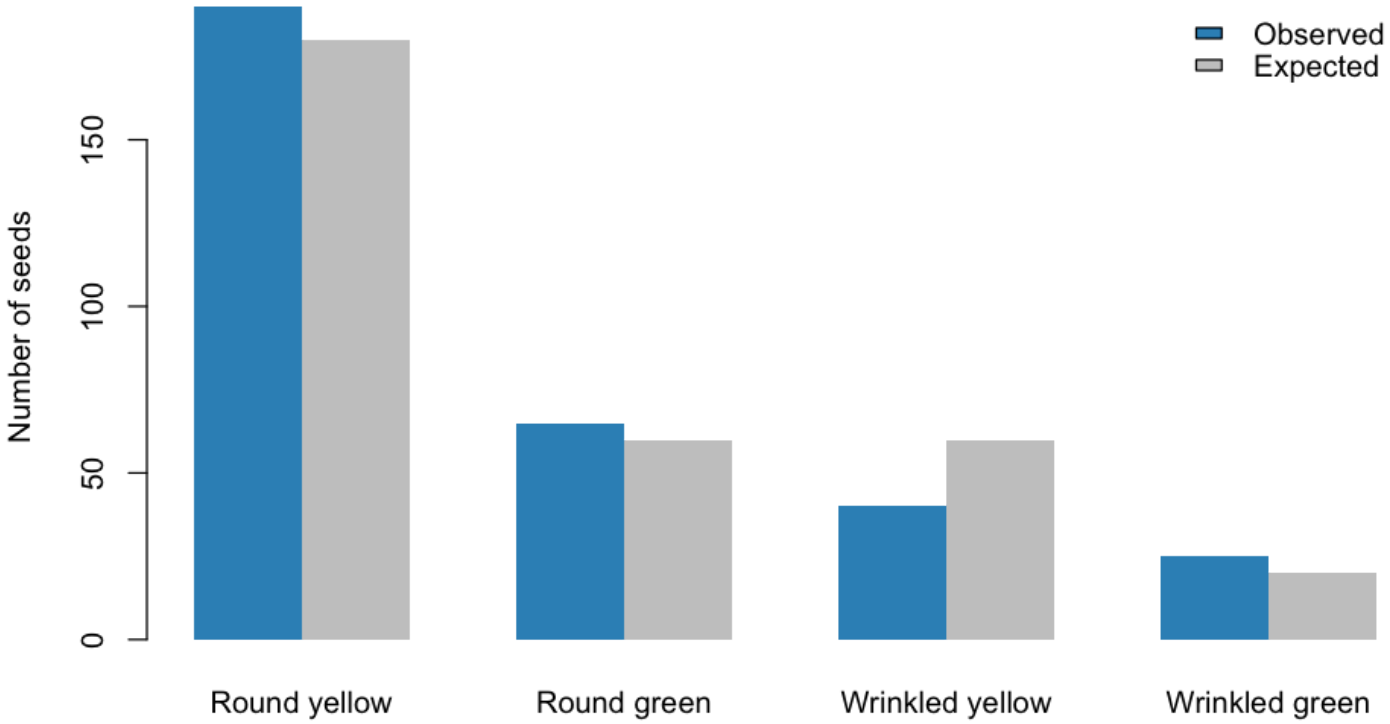
```
    Chi-squared test for given probabilities

data:  observed
X-squared = 8.8889, df = 3, p-value = 0.03081
```

- $\chi^2 = 8.89$ with $df = 4 - 1 = 3$
- $p = 0.03$, which is less than 0.05

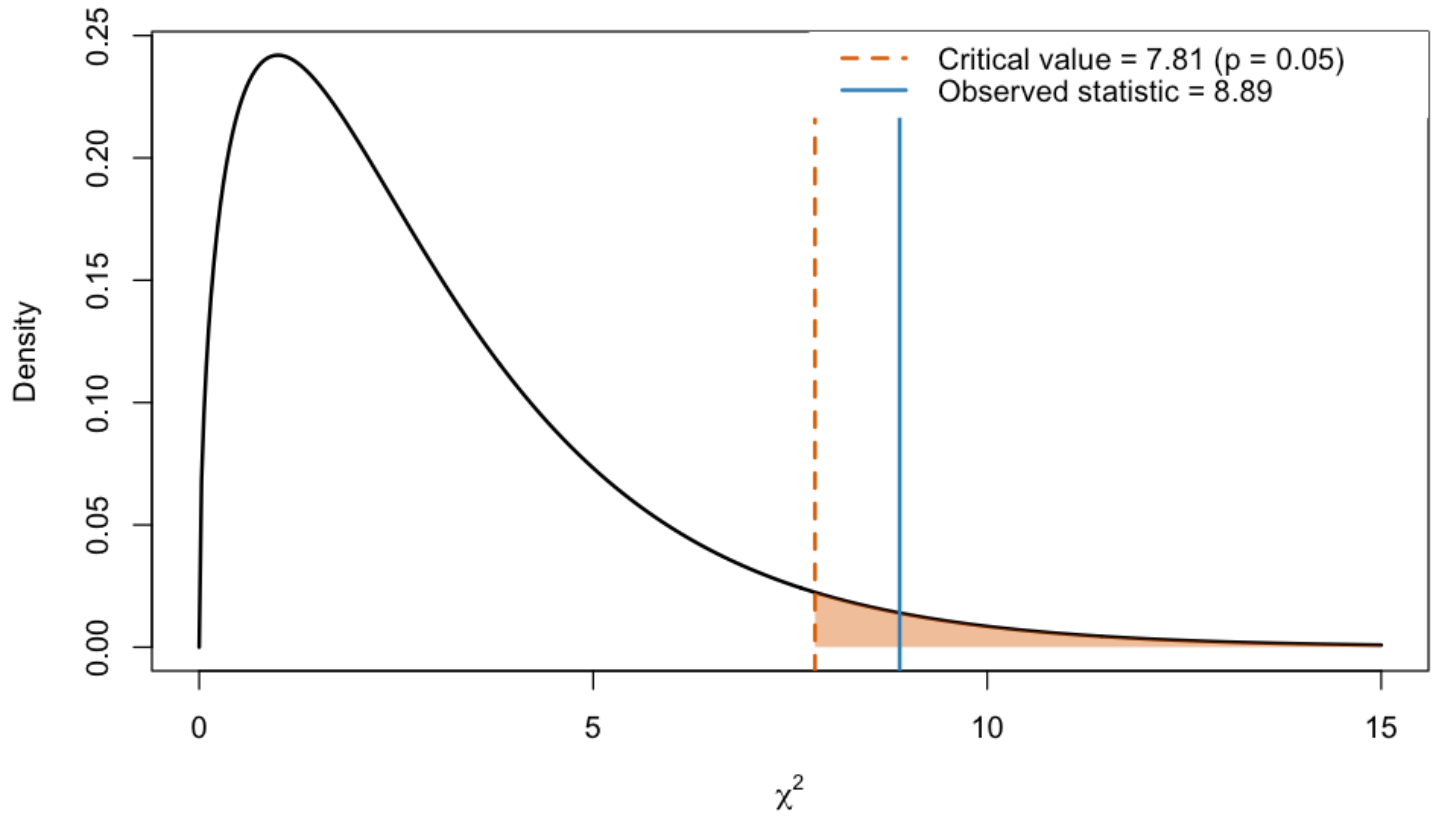
The lab manual reports 8.90 because each term was rounded before adding. Rounding only at the end gives 8.89; the conclusion is the same.

Observed vs. expected



Wrinkled yellow seeds (40 observed vs. 60 expected) account for most of the χ^2 value.

Where does our result fall?



$\chi^2 = 8.89 > 7.81$, so we **reject** H_0: the data do not fit a 9:3:3:1 ratio.

Human connection: MN blood groups

- The M and N blood-group alleles are **codominant**, so all three genotypes (MM, MN, NN) can be identified from a blood test
- Under **Hardy–Weinberg equilibrium** (Hardy, 1908), genotype frequencies are $p^2 : 2pq : q^2$, where p and q are the allele frequencies
- Data: genotypes of 1,279 people in England (Race & Sanger, 1975)

Expected MN genotypes

```r
observed <- c(MM = 363, MN = 634, NN = 282)
n <- sum(observed)
p <- unname((2 * observed["MM"] + observed["MN"]) / (2 * n))  # frequency of M
q <- 1 - p                                                    # frequency of N
expected <- n * c(MM = p^2, MN = 2 * p * q, NN = q^2)
round(c(p = p, q = q), 4)
data.frame(observed, expected = round(expected, 1),
           contribution = round((observed - expected)^2 / expected, 3))
```

p: 0.5317 **q:** 0.4683

A data.frame: 3 × 3

	observed	expected	contribution
	<dbl>	<dbl>	<dbl>
MM	363	361.5	0.006
MN	634	636.9	0.014
NN	282	280.5	0.008

Human connection: result

```r
# df = 3 genotypes − 1 − 1 allele frequency estimated from the data = 1
stat <- sum((observed − expected)^2 / expected)
round(c(chi_square = stat, df = 1,
        p_value = pchisq(stat, df = 1, lower.tail = FALSE)), 3)
```

chi_square: 0.027 **df:** 1 **p_value:** 0.869

- $df = 3 - 1 - 1 = 1$, because one extra value (p) was **estimated from the data**
- $\chi^2 = 0.027 < 3.84$, $p = 0.87$: we **fail to reject** H_0

The population's MN genotypes are consistent with Hardy–Weinberg equilibrium.

Limitations

- χ^2 needs **counts**, not percentages or measurements
- Each expected count should be **at least 5**
- Failing to reject H_0 does **not prove** the hypothesis is true
- A fit can even be *too* good: Fisher (1936) argued that Mendel's published data matched expectations more closely than chance would predict

Conclusion

- Genetics predicts **expected ratios**; real data vary by chance
- χ^2 tells us if the variation is larger than chance would produce
- Compare χ^2 to the critical value and state a conclusion about H_0
- The same test works for plants and human populations

Your turn: apply the test to the corn-ear counts you collected in Lab 11.

References

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Hardy, G. H. (1908). Mendelian proportions in a mixed population. *Science, 28*(706), 49–50. https://doi.org/10.1126/science.28.706.49

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Pearson, K. (1900). On the criterion that a given system of deviations from the probable in the case of a correlated system of variables is such that it can be reasonably supposed to have arisen from random sampling. *Philosophical Magazine, Series 5, 50*(302), 157–175. https://doi.org/10.1080/14786440009463897

Race, R. R., & Sanger, R. (1975). *Blood groups in man* (6th ed.). Blackwell Scientific.

R analysis: chisq.ipynb